

4E1306**4E1306**

B.Tech. IV-Sem. (Main/Back) Examination, July - 2023
Computer Science and Engineering
4CS4-06 Theory of Computation
CS, IT, AID, CAI

Time : 3 Hours**Maximum Marks : 70****Instructions to Candidates:**

Attempt all Ten questions from Part A, Attempt any Five questions out of Seven questions from Part B and Three questions out of Five questions from Part C.

Schematic diagrams must be shown wherever necessary. Any data missing may suitably be assumed and stated clearly. Units of quantities used/calculated must be stated clearly.

Use of following supporting material is permitted during examination (As Mentioned in form No. 205).

PART - A
(Word limit 25)

All questions are compulsory.

(10×2=20)

1. Give the mathematical definition of finite automaton.
2. Construct an NFA, with the specified number of states, that accepts the language $\{w : w \text{ ends with } 10\}$ with three states.
3. Write a regular expressions over $\{0,1\}$ consisting of strings that contain exactly two 1's.
4. Prove $(1+00^*1)+(1+00^*1)(0+10^*1)^*(0+10^*1)=0^*1(0+10^*1)^*$.
5. Explain why the grammar given below is ambiguous.
$$S \rightarrow 0A|1B, A \rightarrow 0AA|1S|1, B \rightarrow 1BB|0S|0$$
6. What is difference between Finite State Automaton and Pushdown Automaton?
7. Explain the Chomsky Hierarchy in details.
8. Can all computational problems solved by computer?
9. What is Halting Problem?
10. List the problems belonging to polynomial class.

PART - B

(Word limit 100)

Attempt any five questions

(5×4=20)

1. Explain following joins with help of example -
 - i. Theta join.
 - ii. Equi Join.
 - iii. Natural join.
 - iv. Outer join.
2. How does triggers are useful in DBMS? Write a trigger in SQL to confirm value inserted in age field of a table is not less than 18 before inserting value.
3. Explain the role and importance of relational algebra. Also explain six basic operators of relational algebra with example.
4. Explain all six inference rules for functional dependency with example.
5. Consider the following schedule of three transactions T1, T2, T3 where X and Y are data items.

T1	T2	T3
	R(X)	
		R(X)
W(Y)		
	W(X)	
		R(Y)
	W(Y)	

Using precedence graph, find out whether the schedule is conflict serializable or not?

6. Explain irrecoverable and recoverable schedules with example.
7. Explain following keys with example -
 - a. Primary key
 - b. Candidate key.
 - c. Super key
 - d. Foreign key.

PART - C

Attempt any Three questions

(3×10=30)

1. How does various E-R model constructs relate to relation model construct. Draw the detailed E-R model for library management system and then convert this ER model to relational model by mapping various constructs.

2. Consider the relations defined below :

DOCTOR (regno, name, telno, city, specialization).

PATIENT (pname, street, city)

VISIT (pname, regno, data_of_visit, fee)

Where the regno and pname fields identify the doctor and the patient uniquely respectively. Express following queries in SQL -

- a. Get the name and regno of doctors who live in Kota. (1)
 - b. Find the name and city of patient(s) who visited a doctor on date 12 - Aug - 2022. (1)
 - c. Find out all doctors whose name start with letter 'n'. (1)
 - d. Find doctors whose specialization in 'NEURO'. (1)
 - e. Find out total number of patients visited by each doctor. (2)
 - f. Print patient name and doctor name to whom he has visited for treatment. (2)
 - g. Find out name of doctors who have not visited any patient yet. (2)
3. Explain Shadow paging and log based recovery techniques with advantages and disadvantages of each.
4. What is normalization? Explain all types of normal forms with example?
5. Write short note on following concurrency control schemes :
- a. Lock - based protocol.
 - b. Timestamp based protocol.